

^{90}Y β^- decay (64.05 h) 1976Gr16,1990Zh20

Type	Author	History	Citation	Literature Cutoff Date
Full Evaluation	S. K. Basu, E. A. Mccutchan		NDS 165,1 (2020)	1-Mar-2020

Parent: ^{90}Y : $E=0$; $J^\pi=2^-$; $T_{1/2}=64.05$ h 5; $Q(\beta^-)=2278.5$ 16; $\% \beta^-$ decay=100

1976Gr16: $\text{Y}(n,\gamma)$, chemical separation. $4\pi\beta$ -counting, $\gamma(t)$ with Ge(Li).

1990Zh20: ^{90}Sr source with ^{90}Y in equilibrium. Measured I_γ with HPGe detector in low-background setup.

For relative measurement of the longitudinal electron polarization, see 1971Jo17.

For studies of shape factors, see 1964Da16, 1966Ri01, 1971Na09, 1975Fi07, 1983Ha35, 2005Gr41.

For measurements of β^- using liquid scintillator detectors see 1993Gr09, 1993Gr18, 1994Gr21, 1994Gr30, 1995Gr04, 2005Gr41;

using solid scintillator detectors, see 1993Va11, 1993Ya17, 1994Us01, 1994Sz09; detecting Cerenkov radiation, see

1994Br45, 1995Ra27; using gas-flow proportional counters, see 1996Be48; using photomultipliers, see 1992Bo35; using ion-chambers, see 1996Wo08.

α : Additional information 1.

 ^{90}Zr Levels

E(level) [†]	J^π	$T_{1/2}$
0	0^+	stable
1760.74 14	0^+	
2186.273 14	2^+	

[†] From the Adopted Levels.

 β^- radiations

E(decay)	E(level)	$I\beta^-$ [†]	Log ft	Comments
(92.2 19)	2186.273	1.4×10^{-6} 3	11.04 10	av $E\beta=24.23$ 45 $I\beta^-$: From measured intensity of 2186 γ (1976Gr16).
(517.8 19)	1760.74	0.0115 14	9.64 ^{1u} 6	av $E\beta=184.59$ 61 $I\beta^-$: From measured $I(\gamma^\pm)/I(\beta^-)=0.000034$ 4 (1961La07) and $I(\text{ce})/I(\gamma^\pm)=2.38$ 8 (1962Ne02).
(2278.5 22)	0	99.9885 14	9.2240 ^{1u} 23	av $E\beta=932.40$ 73 Measured β^- endpoint energies: 2265 keV 5 (1957Yu06); 2261 keV 3 (1958Jo33); 2271 keV 2 (1961Ni02); 2268 keV 2 (1964An12); 2273 keV 5 (1964La13); 2284 keV 5 (1964Da16); 2275 keV 5 (1966Ha15); 2280 keV 5 (1966Ri01); 2275 keV 3 (1993Gr17).

[†] Absolute intensity per 100 decays.

 $\gamma(^{90}\text{Zr})$

Probability of two-photon decay for 1760.7, 0^+ to 0^+ transition is 0.040% 5, weighted av of 0.051% 25 (1973Na01), 0.0317% 77 (1973As04), 0.039% 12 (1989Mu12), and 0.052% 10 (1993Ba65). Others: 1961La07, 1970Ha27, 1970Va09.

Probability of one-photon E0 transition for 1760.7 relative to internal conversion is 5×10^{-7} 2 (1990Zh20). Probability of two-photon E0 transition relative to internal pair conversion is 7.4×10^{-4} 14 (1993Ba65).

E_γ [†]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	$I_{(\gamma+\text{ce})}$ [‡]	Comments
1760.70 20	1760.74	0^+	0	0^+	E0	0.0115 14	ce(K)/(γ +ce)=0.89 1; ce(L)/(γ +ce)=0.11 1 $I_{(\gamma+\text{ce})}$: derived from $I\beta^-$ to this level, see comment on $I\beta^-$. Probability of one-phonon E0 transition relative to internal

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$^{90}\text{Y} \beta^-$ decay (64.05 h) [1976Gr16](#),[1990Zh20](#) (continued) $\gamma(^{90}\text{Zr})$ (continued)

E_γ [†]	I_γ [‡]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	α	Comments
								conversion is 5×10^{-7} 2 (1990Zh20). internal pair production of 1760.7 E0 transition, branching ratio=0.00320 15 pairs/100 decays (2020Pi01), 0.00326 4 pairs/100 decays (2020Dr01), 0.003186 47 pairs/100 decays (2007Se01).
2186.242 25	1.4×10^{-6} 3	2186.273	2 ⁺	0	0 ⁺	E2	5.36×10^{-4}	$\alpha(\text{K})=0.0001223$ 18; $\alpha(\text{L})=1.325 \times 10^{-5}$ 19; $\alpha(\text{M})=2.29 \times 10^{-6}$ 4; $\alpha(\text{N})=3.27 \times 10^{-7}$ 5; $\alpha(\text{O})=2.34 \times 10^{-8}$ 4 I_γ : From absolute counting (1976Gr16). Other: 1.12×10^{-6} 6 from measurement of activity in equilibrium with a ^{90}Sr source (1990Zh20). The method of normalization of 1990Zh20 is however not described.

[†] From the Adopted Gammas.[‡] Absolute intensity per 100 decays.

⁹⁰Y β⁻ decay (64.05 h) 1976Gr16,1990Zh20

Decay Scheme

Intensities: I_(γ+ce) per 100 parent decays

